INFOSYS SPRINGBOARD | DATA VISUALIZATION INTERNSHIP

EDUVISION_DV

Higher Education Intelligence System for Institutional Performance and Academic Excellence Analysis

FINAL PROJECT DOCUMENTATION

Module 8 | Documentation and Project Delivery

Document details	Information
Project	EduVision_DV - Higher Education Performance Dashboard
Prepared by	Nandamuri Pardhavi
Internship role	Data Visualization Intern
Platform	Infosys Springboard
Project tools	Python, Pandas, NumPy, Tableau, Git and GitHub
Document version	1.0 | 15 October 2026
Delivery status	Documentation prepared; final GitHub push to be verified

This report describes the implementation and deliverables documented to date. Technical details that require inspection of the final code or workbook are clearly identified.

Document Control and Reading Guide

Item	Description
Purpose	Consolidated technical and delivery report for Modules 1-8
Intended readers	Internship mentor, evaluator, academic reviewers and portfolio visitors
Primary evidence	Milestone scripts, datasets, Tableau packaged workbook and Module 7 QA reports
Verification convention	“Completed” refers to user-confirmed work or files visible in the supplied Git staging output; “verify” flags details not independently inspected.

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01 Executive Summary

EduVision_DV is a higher-education analytics project designed to convert heterogeneous university-ranking and education-statistics data into an accessible, interactive Tableau dashboard suite. The project connects institutional performance, research, student-related indicators and country-level comparisons so that users can move from raw records to interpretable education insights.

The work was developed through eight modules grouped into four milestones: collection and cleaning; KPI engineering and dashboard planning; Tableau development and integration; and testing, documentation and delivery. Python scripts and notebooks support data preparation; calculated education indicators form an analytical layer; Tableau provides four thematic dashboards; and GitHub supports version-controlled handover.

Area	Implemented or available output
Data preparation	Source datasets, collection script, cleaning notebook and cleaned workbooks
Analytical layer	Six KPI workbook and university-level final dataset
Visualization	Four completed dashboards; packaged Tableau workbook in Modules 5 and 6
Quality assurance	Dashboard_Testing_Report.pdf and QA_Checklist.pdf in Module 7
Delivery	Existing mentor repository and Nandamuri_Pardhavi branch; final push of latest staged changes remains to be confirmed

02 Problem Statement and Objectives

Problem statement

University performance information is distributed across multiple ranking systems and country-level education datasets. Differences in coverage, naming, years, metrics and data quality make direct comparison difficult. A single ranking position does not explain research strength, academic reputation, international participation or the broader education context.

Project objectives

• Collect and document relevant university and education-statistics datasets.

• Clean, standardize and validate university, country and indicator fields.

• Develop six meaningful education KPIs for comparison and interpretation.

• Design and integrate four interactive Tableau dashboards.

• Enable filtering, navigation and comparative analysis for different users.

• Validate calculations and dashboard interactions; deliver reusable documentation and project artifacts.

Intended users

Students and applicants exploring universities; academic researchers comparing institutions; university administrators reviewing performance; education consultants and policymakers examining institutional and country-level patterns.

03 Project Scope and Expectations

Manual expectation	Expected evidence / outcome
Module 1 - Collection	Datasets integrated; source inventory and completeness checks (manual target above 95%).
Module 2 - Cleaning	Consistent, analysis-ready fields; duplicate, null and type checks.
Module 3 - KPI Engineering	Six documented KPIs and final analytical dataset.
Module 4 - Design	Storyboard, layout specifications and dashboard prototype.
Modules 5-6 - Tableau	Four dashboards with coordinated filters, interactions and navigation.
Module 7 - QA	Testing checklist and documented validation findings.
Module 8 - Delivery	GitHub repository, final documentation and Tableau workbook; portfolio-ready presentation.

A target is not automatically a measured result. For example, the >95% completeness requirement is an acceptance target; its actual achieved percentage should be copied from the final validation report or script output before claiming it was met.

04 Technology and Architecture

Technology	Role in the project
Python / Pandas	Dataset loading, joins, standardization, cleaning and KPI engineering
NumPy	Numerical operations where required by scripts
Jupyter notebook	Interactive cleaning, transformation and exploratory checks
Excel / CSV	Source, intermediate, KPI and final tabular datasets
Tableau Desktop	Interactive charts, dashboard integration and packaged workbook
Git / Git LFS / GitHub	Version history, large-data handling and repository-based delivery

Logical data flow

QS 2025 / THE 2024 / WUR 2023 / World Bank EdStats → source validation → cleaning and standardization → thematic datasets → KPI calculations and university matching → Tableau data connections → four integrated dashboards → QA evidence → final documentation and GitHub delivery.

Institution-level rankings and country/year education indicators have different analytical grains. The project should not treat country-level records as if they were individual universities. University joins require carefully normalized identifiers and explicit checks for unmatched records.

05 Dataset Sources and Data Preparation

Source dataset	Coverage / primary role	Known local file
QS World University Rankings 2025	Global university rankings, academic reputation, institutional indicators	QS World University Rankings 2025 (Top global universities).csv
Times Higher Education 2024	Teaching, research and international-performance indicators	TIMES_WorldUniversityRankings_2024.csv
World University Rankings 2023	Student population, student-staff measures and research-related attributes	World University Rankings 2023.csv
World Bank EdStats	Country/year education context and international education indicators	EdStatsData.csv and related EdStats reference files

Dataset names and years are taken from the project file inventory. Original publication/download URLs and licenses should be added from the actual source metadata before external redistribution. The World Bank source is not a university-level ranking file.

Collection and source validation

Module 1 uses data_collection.py and raw datasets to establish source availability and an initial integration foundation. Previously recorded collection outputs included 1,503 QS rows and 2,673 THE rows. These are stage-specific counts, not a claim about the final dashboard record count. Source quality checks include schema availability, missing values, duplicates, valid types and integration completeness.

Cleaning and transformation

Module 2 uses education_cleaning.ipynb and produces four themed Excel files: 01 University Overview.xlsx, 02 Research Analytics.xlsx, 03 Student Analytics.xlsx and 04 Country Comparison.xlsx. The pipeline addresses inconsistent university/country naming, numeric fields, missing data, duplicates and reusable output schemas.

Data issue	Handling / verification principle
Different university spellings	Normalize display/matching keys; inspect uncertain matches manually.
Missing indicator values	Record null counts; avoid silently replacing unavailable metrics with invented values.
Duplicated university records	Check duplicate keys and select a defensible deduplication rule.
Mixed numeric formats	Convert ranks, scores and percentages with explicit coercion checks.
Year mismatch	Preserve dataset vintage; avoid presenting cross-year comparisons as same-year trends.

06 Milestone-wise Implementation and Outputs

Module	Work performed / implementation	Evidence / output
1. Data Collection	Gathered QS, THE, WUR and EdStats files; prepared collection script and raw source folder.	data_collection.py; raw CSV files
2. Data Cleaning	Prepared cleaned thematic workbooks using a notebook-based workflow.	education_cleaning.ipynb; four Excel files; university_cleaned.csv
3. KPI Engineering	Developed Python KPI processing and a university-level analytical dataset.	generate_education_kpis.py; Six KPIs.xlsx; university_final_dataset.xlsx
4. Dashboard Planning	Defined four dashboard topics and intended KPI, chart, filter and navigation placements.	dashboard_storyboard.pdf; dashboard_layout_spec.md (previously reported)
5. Dashboard Development	Built Tableau dashboard views for ranking, research, students and country comparison.	EduVision_DV.twbx in Module_05
6. Integration	Completed four-dashboard suite and integrated workbook for navigation and analysis.	EduVision_DV.twbx in Module_06
7. Testing	Prepared dashboard testing and QA deliverables.	Dashboard_Testing_Report.pdf ; QA_Checklist.pdf
8. Delivery	Consolidating project documentation and preparing repository handover.	This PDF; existing GitHub branch; push pending verification

Practical challenges encountered

During KPI integration, university identifiers from different workbooks did not share a common format. An initial join by university_id returned no matching rows. The workflow was revised to use standardized university-name keys, with a previously observed 917 matches out of 918 KPI records and one unmatched institution requiring review. This illustrates why semantic key matching and exception reporting matter more than simply forcing a high join rate.

The final dataset also previously showed missing values in several ranking-related fields (for example, overall_score and selected international metrics). These should remain visible in data-quality reporting; completeness must be evaluated by field and use case, not assumed to be universal.

07 KPI Engineering and Definitions

Module 3 establishes six named KPIs. Their meanings and intended interpretations are documented below. The exact executable formulas, weighting, fallback logic and normalization boundaries must be transcribed from the final generate_education_kpis.py before treating this as a formula-level specification.

KPI	Business meaning	Interpretation / caution
Global Ranking Score	Summary measure of overall institutional ranking performance.	Higher score usually indicates stronger performance; check source scale.
Academic Reputation Score	Perceived academic standing based on the selected ranking indicators.	Reputation measure is source-dependent.
Research Impact Score	Measure of research influence or citation/research performance.	Not identical to publication volume.
Faculty-to-Student Ratio	Relative teaching staff availability per student.	Confirm numerator/denominator and scale.
International Student Percentage	Share of international students in the relevant student population.	Confirm whether input is already a percentage.
Research Productivity Index	Composite or derived indicator of research output relative to a defined base.	Document precise formula and source fields before comparing institutions.

KPI processing and integration

The KPI script reads thematic Module 2 workbooks and writes Six KPIs.xlsx and university_final_dataset.xlsx. The six indicators form a common comparison layer for the dashboards. For auditability, the final code should specify source columns, transformations, score directions, treatment of zero denominators, normalization, rounding and missing-value policy.

Required formula register for final sign-off

Field to verify	Where to verify
Exact Python expression for each of the six KPIs	Milestone_02/Module_03/scripts/generate_education_kpis.py
Input columns, units and score range	KPI script and Six KPIs.xlsx
Number of output rows and unmatched keys	Latest script run log and university_final_dataset.xlsx
Null count and validation results	Latest dataset profile / QA report

08 Tableau Dashboard Suite and User Guide

The final visualization package is organized into four dashboards. These sections describe the intended analytical experience and the user-confirmed completed dashboard suite; exact worksheet titles and control labels should be verified against the packaged Tableau workbook.

University Overview

Analytical focus: Global ranking performance, academic reputation and institutional comparison.

How to use: Select a country or university, review overall ranking context, and compare selected institutions across available KPIs.

Expected output: Ranking comparisons and a consolidated institutional-performance view.

Research Analytics

Analytical focus: Research impact, productivity and related research-performance indicators.

How to use: Choose institutions or regions and inspect how research-oriented indicators differ across peers.

Expected output: Comparative research profile and impact interpretation.

Student Analytics

Analytical focus: International participation, student-related metrics and staff-to-student context.

How to use: Filter institutions and compare student composition and available ratios.

Expected output: Student diversity and resource-comparison view.

Country Comparison

Analytical focus: Cross-country and regional educational comparisons using compatible country-level indicators.

How to use: Choose countries/regions and compare available measures; interpret years and units carefully.

Expected output: Country-level benchmark and contextual education view.

General dashboard operating guide

- Open the packaged workbook EduVision_DV.twbx in a compatible Tableau environment.
- Use the landing/navigation controls or workbook dashboard tabs to select a thematic dashboard.
- Apply the available country, university, region or year filters as appropriate to the sheet.
- Use hover tooltips, legends and selections to inspect values and make comparisons.
- Clear selections or reset filters before starting a different comparison.
- Check source year and metric definition when comparing data from different ranking systems.

Tableau Public publication is optional in the manual. No public Tableau URL has been confirmed for this report.

09 Analytics Methodology and Data Quality

Methodological stages

Stage	Procedure	Quality gate
Source inventory	Identify source, year, grain, schema and licensing.	Dataset traceability
Schema alignment	Standardize names, numeric types and country identifiers.	Comparable field definitions
Matching	Normalize university names; inspect exceptions.	No unsupported false matches
KPI construction	Compute six defined indicators using script logic.	Reproducible calculations
Visualization	Map indicators to dashboards with meaningful filters.	Consistent aggregation and labeling
Validation	Check metrics, navigation, filters and dataset coverage.	Recorded QA results

Interpretation safeguards

Ranking systems use different methodologies and years. Their scores are not necessarily interchangeable, and changes between datasets should not automatically be interpreted as longitudinal improvements. Aggregation should respect metric units; for example, ratios and percentages often require weighted treatment rather than summation. Country-level indicators should be compared at a compatible year and geographic scope.

Data-quality indicators

The project manual specifies a completeness target above 95% for the relevant collection milestone. Separately report missingness, duplicate rates, source-to-target coverage, match rate and KPI validity. A university match rate and a dataset completeness percentage are different measurements and should never be conflated.

10 Testing, Validation and Acceptance

Module 7 contains two submitted QA artifacts: Dashboard_Testing_Report.pdf and QA_Checklist.pdf. Their presence is confirmed from the local Git staging output. The individual pass/fail entries and numerical outcomes have not been independently inspected in preparing this report.

Test area	Expected acceptance behavior	Evidence to review
Data source connections	Workbook loads intended datasets without broken links.	Open packaged workbook
KPI accuracy	Displayed values agree with validated analytical dataset.	Script outputs and dashboard samples
Filters and navigation	Selections update intended views; reset behavior is predictable.	QA checklist and manual test

Test area	Expected acceptance behavior	Evidence to review
Ranking / country comparisons	Correct grouping, labels, units and sort order.	Dashboard screenshots and report
Missing values	Unavailable data is handled transparently.	Dataset profile and tooltips
Usability	Readable titles, legends and clear interaction paths.	Final dashboard review

Acceptance status

The user has confirmed that all four dashboards are complete and that the Module 7 report and checklist have been produced. A formal “all tests passed” statement should be made only after reviewing the contents of those QA files and resolving any open issues.

11 Module 8 Delivery and GitHub Structure

Required Module 8 deliverables

Manual requirement	Project artifact / status
GitHub Repository	Existing mentor repository and Nandamuri_Pardhavi branch; latest milestone push pending confirmation.
Final Documentation	This consolidated PDF.
Tableau Workbook	EduVision_DV.twbx in Milestone_03/Module_06/deliverables.
Portfolio-ready suite	Four completed thematic dashboards; README and deployment verification still to be finalized.

Repository reference

Mentor repository: github.com/springboardmentor1234r/Higher-Education-Intelligence-System-for-Institutional-Performance-and-Academic-Excellence-Analysis

Working branch: Nandamuri_Pardhavi. The latest observed Git status showed Milestones 1-4 staged, but not committed or pushed. Therefore this document does not claim that the new versions are already visible remotely.

Recommended project navigation

Retain the established Milestone_01 through Milestone_04 hierarchy. In the root README, add a concise navigation table linking readers to scripts, data, dashboard and docs locations. Module 8 deliverables should live under Milestone_04/Module_08/deliverables. The manual’s /scripts, /data, /dashboard and /docs categories can be satisfied through a clear README map without moving all milestone artifacts.

Large file management

EdStatsData.csv was reported at 326,426,040 bytes. GitHub regular Git files above 100 MB are rejected; this dataset should be verified as an LFS pointer at its current path before pushing. The earlier branch used Git LFS for a previous EdStatsData.csv path, which does not itself verify the new staged location.

12 Results, Limitations and Future Work

Project outputs

Output	Value delivered
Curated educational data	Thematic, reusable university and country datasets.
KPI layer	Six education-focused indicators for comparison.
Four Tableau dashboards	Interactive overview, research, student and country analytics.
Quality artifacts	Testing report and QA checklist for handover.
Project handover	Packaged workbook, documentation and version-controlled repository structure.

Known limitations

• Source datasets cover different ranking years and do not represent one synchronized census.

• Cross-source university-name matching can leave unresolved institutions; uncertain matches should not be forced.

• Some source fields contain missing values, limiting complete comparisons.

• KPI formula details and final QA pass counts require inspection of the current source files.

• The final branch push and optional Tableau Public publication have not been verified.

Future enhancements

Potential improvements include automated dataset refreshes, source metadata tables, explicit data lineage, regression tests for KPI calculations, more robust entity resolution, interactive year comparisons where compatible data exists, and a public portfolio landing page linking the README, workbook and final report. These are recommendations, not completed features.

13 Appendix: Deliverable Inventory and Final Checklist

Artifact	Project location / description
Collection script	Milestone_01/Module_01/scripts/data_collection.py
Cleaning notebook	Milestone_01/Module_02/scripts/education_cleaning.ipynb
Cleaned workbooks	Milestone_01/Module_02/deliverables/university_cleaned/
KPI script	Milestone_02/Module_03/scripts/generate_education_kpis.py
KPI workbook	Milestone_02/Module_03/Six KPIs.xlsx
Final analytical dataset	Milestone_02/Module_03/deliverables/university_final_dataset.xlsx
Tableau workbook	Milestone_03/Module_06/deliverables/EduVision_DV.twbx
QA reports	Milestone_04/Module_07/deliverables/
Final documentation	Milestone_04/Module_08/deliverables/Final_Project_Documentation.pdf

Final submission checklist

• Confirm the final KPI formulas and dataset counts against the current script outputs.

• Verify Module 7 QA outcomes and insert actual pass/fail summary.

• Review README links and dataset-source citations/licenses.

• Verify Git LFS for large files and push intended milestone changes.

• Open the Tableau workbook from the final delivery package.

• Commit the Module 8 PDF and verify its GitHub branch URL.

Closing statement. EduVision_DV demonstrates a complete practical workflow from multi-source educational data preparation through KPI engineering, interactive visualization and documented delivery. The final handover should prioritize traceability, accurate metric definitions, clear dashboard operation and reproducibility rather than unsupported performance claims.